1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=3;i++)
4       (i-1)[arr] = (i%3);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 1;
8   }
9  int main(){
10   int arr[4] = {0,0,0,0};
11   func1(arr+1);
12   for(int i=0; i<4; i++)
13          printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<4; i++)
17          printf("%d ", i[arr] );
18   return 0;
19 }
```

arr [i-1] = i%3
arr[0] = 1
arr[1] = 2
arr[3] = 0

{0, 1, 2, 0}

func2 (arr, 0)

Output
0 1 2 0
1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 1;      0
  a = &temp;
}
```

Explain your answer

Yes, as now a new memory address is formed that is &temp

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

40 x 8
40 x 4

a_1 and a_2 stores address of array of long int type while b_1 and b_2 store address of of int type.

Output: 240 160

160 = (50 − 10) × 4 4 = space of int type

240 = (50 − 10) × 6 6 = space of long type